Stat 266 – Unit 10 – Video Set 2 – Class Examples

Summary:

2 **Large Sample (*Z-Tools*) Tests**

- Parameter: θ
- Consider testing $H_0 : \theta = \theta_0$ vs. $H_A : \begin{cases} i) & \theta > \theta_0 \\ ii) & \theta < \theta_0 \\ iii) & \theta \neq \theta_0 \end{cases}$
- Sampling Distribution of $\hat{\theta}$: If $\hat{\theta} \approx Normal(\underbrace{\mu_{\hat{\theta}} = \theta}_{\text{unbiased}}, \sigma^2_{\hat{\theta}})$, then

$$\frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \approx Normal(0,1).$$

- Test-Statistic: $TS = z_0^* = \dfrac{\hat{\theta} - \overset{\text{under } H_0}{\theta_0}}{\hat{\sigma}_{\hat{\theta}}}$.
- $RR_\alpha = \begin{cases} i) & \{z_0^* > z_\alpha^*\} \\ ii) & \{z_0^* < -z_\alpha^*\} \\ iii) & \{|z_0^*| > z_{\alpha/2}^*\} \end{cases}$, $\quad p-value = \begin{cases} i) & P(Z > z_0^*) \\ ii) & P(Z < z_0^*) \\ iii) & 2P(Z > |z_0^*|) \end{cases}$

1 **Power of the Test:**

Consider some value of θ under H_a, θ_a. Then the power of the test is defined as

$$* \Rightarrow power(\theta_a) = P(\text{Reject } H_0 \mid \underbrace{\theta = \theta_a}_{H_0 \text{ is false}}) = P(\hat{\theta} \in RR \mid \underbrace{\theta = \theta_a}_{H_0 \text{ is false}}) = 1 - \overset{P(\text{type II error})}{\beta_{\theta_a}}.$$

$= P(\text{fail to reject } H_0 \mid H_0 \text{ is false})$

Examples:

1) Parameter: $\theta = \mu_Y$

$$TS = Z_0^* = \frac{\bar{y} - \mu_0}{\sigma_Y / \sqrt{n}} \approx \frac{\bar{y} - \mu_0}{S_Y / \sqrt{n}}$$

↳ large n

OR

$$\frac{\bar{y} - \mu_0}{S_Y / \sqrt{n}} \sim t_{n-1}$$

2) Parameter: $\theta = p$

$H_0 : p = p_0$

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{p_0(1-p_0)}{n}}$$

↳ under H_0

$$TS = Z_0^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}.$$

Example C : Suppose $Y_1,\ldots,Y_n$ is a random sample of lifetimes of lightbulbs (in hundreds of hours). Suppose also that all bulbs are independent with lifetime described by the following probability density:

$$\Rightarrow f_Y(y)=\begin{cases}\frac{1}{\theta}e^{-y/\theta} & , y\geq 0\\ 0 & , \text{else}\end{cases}$$

$Y \sim Exp(\theta = \mu_Y)$

Parameter: population mean

The company that manufactures the bulbs advertise that average lifetime is 300 hundred hours. A consumer group believes otherwise that average lifetime is under 300 hours.

The test set-up is: $H_0:\theta=3 \quad vs. \quad H_a:\theta<3$

Based on n observations the consumer group has decided to reject the null hypothesis based on the following rejection region (rule): $RR=\{\bar{Y}:\bar{y}<k\}$.

Define the power of the test at $\theta=\theta_a<3$.

a) If $n=35$ and $\alpha=0.10$, what is $power(\theta_a=2.75)$?

b) If $n=50$ and $\alpha=0.10$, what is $power(\theta_a=2.00)$?

Recall: $\frac{2n\bar{Y}}{\theta} \sim \chi^2_{2n}$

$$power(\theta_a) = P(\underbrace{\bar{Y}<k}_{\text{reject } H_0} \mid \underbrace{\theta=\theta_a}_{H_0 \text{ is false}}) = P\left(\frac{2n\bar{Y}}{\theta} < \frac{2nk}{\theta} \mid \theta=\theta_a\right)$$

$$= P\left(\chi^2_{2n} < \frac{2nk}{\theta_a}\right).$$

Recall: For fixed α, $k = \frac{\theta_0}{2n}\chi^{2*}_{2n,1-\alpha}$.

$$\text{power}(\theta_a) = P\left(\chi^2_{2n} < \frac{2nk}{\theta_a}\right)$$

$$= P\left(\chi^2_{2n} < \frac{2n}{\theta_a} \cdot \frac{\theta_0}{2n} \chi^{2*}_{2n,1-\alpha}\right) = P\left(\chi^2_{2n} < \frac{\theta_0}{\theta_a} \chi^{2*}_{2n,1-\alpha}\right)$$

a) If $n = 35$, $\alpha = 0.10$, and $\theta_0 = 3$:

$$\chi^{2*}_{70,0.90} = 55.329$$

$$\text{power}(\theta_a) = P\left(\chi^2_{70} < \frac{1}{\theta_a}(165.987)\right)$$

e.g. $\text{power}(\theta_a = 2.75) = P(\chi^2_{70} < 60.359)$

$$= 1 - \underbrace{P(\chi^2_{70} > 60.359)}_{\in(0.10,\,0.90)} \in (0.10, 0.90)$$

Actual $= 0.2123$.

b) If $n = 50$, $\alpha = 0.10$, and $\theta_0 = 3$: $\chi^{2*}_{100,0.90} = 82.358$

$$\text{power}(\theta_a) = P\left(\chi^2_{100} < \frac{1}{\theta_a}(247.074)\right).$$

e.g. $\text{power}(\theta_a = 2.00) = P(\chi^2_{100} < 123.537)$

$$= 1 - P(\chi^2_{100} > 123.537) \in (0.90, 0.95)$$

Actual: 0.9447

c) For fixed $\alpha = 0.05$ and power$(\theta_a = 2.90) = 0.95$, what is the minimum sample size needed?

$H_0: \theta = 3$ vs $H_a: \theta < 3$

$$\text{power}(\theta_a = 2.90) = P\left(\chi^2_{2n} < \frac{3}{2.9}\chi^{2*}_{2n,\underbrace{0.95}_{1-\alpha}}\right) \Rightarrow 0.95$$

(trial and error $\Rightarrow$ technology)

$$n = 100: \quad P\left(\chi^2_{200} < \frac{3}{2.9}(168.278)\right) = P\left(\chi^2_{200} < 174.08\right)$$
$$\approx 0.0929$$

$$n = 10000: \quad P\left(\chi^2_{20000} < \frac{3}{2.9}(19672.17)\right) \approx 0.9596$$

$n \approx 9400 - 9500$

Example D : Suppose you select a sample of n observations from a population with probability density

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{3} e^{-\frac{(y-\theta)}{3}} & , \quad \theta < y \\ 0 \ , \ \text{else} \end{cases}$$

θ = population minimum

$$f_{Y_{(1)}}(y) = \frac{n}{3} e^{-\frac{n}{3}(y-\theta)} \quad , \ y > \theta$$

You wish to carry out the following test based on the statistic $Y_{(1)}$.

$$H_0 : \theta = 2 \ vs. \ H_a : \theta > 2$$

The rejection region is $RR = \{Y_{(1)} > k\}$.

a) Suppose $n = 25$. If $power(\theta_a = 2.15) = 0.95$, what is α (the significance level of the test)?

b) If you want $power(\theta_a = 2.15) = 0.95$ for $\alpha = 0.01$, how many observations do you need?

a) $$power(\theta_a) = P(Y_{(1)} > k \mid \theta = \theta_a) = \int_k^\infty \frac{n}{3} e^{-\frac{n}{3}(y-\theta_a)} dy$$

$$= e^{-\frac{n}{3}(k-\theta_a)}.$$

<u>Recall:</u> For fixed α, $k = \theta_0 - \frac{3}{n} \ln \alpha$.

$$\Rightarrow power(\theta_a) = e^{-\frac{n}{3}(\theta_0 - \frac{3}{n}\ln\alpha - \theta_a)} = \alpha e^{-\frac{n}{3}(\theta_0 - \theta_a)}.$$

e.g. If $\theta_0 = 2$, $n = 25$ and $power(\theta_a = 2.15) = 0.95$,

$$0.95 = e^{-\frac{25}{3}(k-2.15)} \Rightarrow k = 2.15 - \frac{3}{25}\ln(0.95) = 2.156155.$$

$$\alpha = e^{-\frac{n}{3}(k-2)} = e^{-\frac{25}{3}(2.156155-2)} \approx e^{-1.30129} \approx 0.27218.$$

OR

$$\text{power}(\theta_a) = \alpha e^{-\frac{n}{3}(\theta_0 - \theta_a)}$$

$$\Rightarrow \alpha = \text{power}(\theta_a) \cdot e^{\frac{n}{3}(\theta_0 - \theta_a)}$$

$$= 0.95 e^{\frac{25}{3}(2 - 2.15)} \approx 0.27218.$$

b) $\text{power}(\theta_a) = \underbrace{\alpha e^{-\frac{n}{3}(\theta_0 - \theta_a)} \Rightarrow 1 - \beta_{\theta_a}}_{\text{solve for } n}$

$$n = \frac{3(\ln(1-\beta) - \ln(\alpha))}{\theta_a - \theta_0}$$

If we want $\text{power}(\theta_a = 2.15) = 0.95$ and $\alpha = 0.01$,

$$n = \frac{3(\ln(0.95) - \ln(0.01))}{2.15 - 2.00} = 91.08 \Rightarrow 92.$$

Example E : Suppose $Y_1, Y_2, \ldots, Y_n$ are iid r.v.'s with p.d.f

$$f_Y(y) = \begin{cases} \dfrac{3\theta^3}{y^4} & , 0 < \theta < y \\ 0 & , else \end{cases}$$

$\Rightarrow f_{Y_{(1)}}(y) = \dfrac{3n\theta^{3n}}{y^{3n+1}}, \quad y > \theta$

You wish to carry out the following test based on the statistic $Y_{(1)}$:

$$H_0 : \theta = 5 \; vs. \; H_a : \theta > 5.$$

Recall: For fixed α, $k = \theta_0(\alpha)^{-1/3n}$.

The rejection region is $RR = \{Y_{(1)} : y_{(1)} > k\}$.

a) What is the power of the test if $\theta = \theta_a = 5.25$, for $\alpha = 0.10$? Suppose $n = 10$.

b) If you want $power(\theta_a = 5.25) = 0.90$ for $\alpha = 0.05$, how many observations do you need?

a) $\text{power}(\theta_a) = P(Y_{(1)} > k \mid \theta = \theta_a) = \displaystyle\int_k^\infty \frac{3n\theta_a^{3n}}{y^{3n+1}}\,dy = \left(\frac{\theta_a}{k}\right)^{3n}$

$$= \frac{\theta_a^{3n}}{\theta_0^{3n}(\alpha)^{-1}} = \alpha\left(\frac{\theta_a}{\theta_0}\right)^{3n}.$$

$\text{power}(\theta_a = 5.25) = 0.10\left(\dfrac{5.25}{5}\right)^{30} = 0.4322.$

Note: what if $n = 50$? $(\alpha = 0.10)$

$\text{power}(\theta_a = 5.25) = 0.10\left(\dfrac{5.25}{5}\right)^{150} = 150.8 > 1$ ✗

$\Rightarrow$ If $n = 50$ and $\alpha = 0.10$, $k = 5(0.10)^{-1/150} = 5.077$.

If $\theta = \theta_a = 5.25$ then $y_{(1)}$ is surely larger than 5.25.

$\Rightarrow$ we will surely reject H_0.

$\Rightarrow \text{power}(\theta_a = 5.25) = 1.$

b) $\text{power}(\theta_a) = \alpha\left(\frac{\theta_a}{\theta_0}\right)^{3n} \Rightarrow 1-\beta_{\theta_a}$

$$\Rightarrow n = \frac{\ln(1-\beta_{\theta_a}) - \ln\alpha}{3(\ln\theta_a - \ln\theta_0)}$$

$$= \frac{\ln(0.90) - \ln(0.05)}{3(\ln(5.25) - \ln(5))} = 19.7 \Rightarrow 20.$$

Example A: An experiment to explore the pain thresholds to electrical shocks for males and females resulted in the data summary below. Suppose the pain thresholds for both males and females are normally distributed with known variance stated in the table below.

	X Males	Y Females
n	14	10
Sample mean	16.2	14.9
Population Variance	12.7	26.4

Parameter:

$\theta = \mu_x - \mu_y$

Is there a difference in population means between males and females? Carry out an appropriate test at α=0.10.

H_0: (no) $\mu_x - \mu_y = 0$ (θ_0), H_a: (yes) $\mu_x - \mu_y \neq 0$

$\hat{\theta} = \bar{x} - \bar{y} = 16.2 - 14.9 = 1.30$

$\sigma^2_{\hat{\theta}} = V[\bar{X} - \bar{Y}] = V[\bar{X}] + V[\bar{Y}] = \frac{\sigma_x^2}{n_1} + \frac{\sigma_y^2}{n_2} = \frac{12.7}{14} + \frac{26.4}{10}$

$= 3.547.$

$\sigma_{\hat{\theta}} = \sqrt{3.547} \approx 1.8834$

$Z^*_0 = \frac{\hat{\theta} - \theta_0}{\sigma_{\hat{\theta}}} = \frac{1.30 - 0}{1.8834} = 0.69$

p-value $= 2P(Z > |Z^*_0|) = 2P(Z > 0.69) = 2(0.2451)$

$= 0.4902$

$> \alpha = 0.10$

With a p-value = 0.4902 we have insignificant evidence of a difference between μ_x and μ_y, at $\alpha = 0.10$.

Consider a $\underbrace{90\%}_{(1-\alpha)100\%}$ CI for $\mu_x - \mu_y$?

$\Rightarrow \alpha = 0.10$

- p-value $= 0.4902 > \alpha = 0.10$

  $\Rightarrow$ fail to reject H_0 (fail to reject "0")

$\Rightarrow$ The 90% CI will include 0.

$\Rightarrow 1.30 \pm \{1.645 \times 1.8834\} \Rightarrow (-1.798, 4.398)$

Example B: A random sample of homeowners and condominium owners were asked whether they planned any landscaping within the next year. The data are given in the following table.

	Residence	Sample size	Number who are Planning to landscape
1	Homeowner	260	91
2	Condominium owner	300	60

Is the population proportion of people who are planning to landscape for homeowners higher than the proportion for condominium owner? Carry out an appropriate test at α=0.01.

Parameter: $\theta = p_1 - p_2$

$H_0: \underbrace{p_1 - p_2 = 0}$, $H_a: p_1 - p_2 > 0$

{ OR $p_1 - p_2 \le 0$ }

$$\hat{\theta} = \hat{p}_1 - \hat{p}_2 = \frac{91}{260} - \frac{60}{300} = 0.35 - 0.20 = 0.15$$

$$\sigma^2_{\hat{\theta}} = V[\hat{p}_1 - \hat{p}_2] = V[\hat{p}_1] + V[\hat{p}_2] = \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$$

{ Note: Under H_0, $p_1 - p_2 = 0 \Rightarrow p_1 = p_2 \Rightarrow p$ }

$$= p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$$

The best estimate for p is

$$\hat{p} = \frac{91+60}{260+300} = \frac{n_1\hat{p}_1 + n_2\hat{p}_2}{n_1+n_2} = \frac{151}{560} = 0.2696.$$

$$\hat{\sigma}^2_{\hat{\theta}} = \hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right) = 0.0014138965$$

$$\hat{\sigma}_{\hat{\theta}} = 0.0376018$$

$$Z_0^* = \frac{\hat{\theta} - \theta_0}{\hat{\sigma}_{\hat{\theta}}} = \frac{0.15 - 0}{0.0376018} = 3.99$$

$$p\text{-value} = P(Z > 3.99) \approx P(Z > 4) \approx 0.0000317$$

$$< \alpha = 0.01$$

With a p-value of 0.0000317 we have significant evidence to suggest $p_1 > p_2$, at $\alpha = 0.01$.

<u>Note:</u> Consider a two-tailed 99% CI for $p_1 - p_2$.

- A two-tailed p-value $= 2(0.0000317) < \alpha = 0.01$.
- Since $\hat{p}_1 - \hat{p}_2 > 0$, the 99% CI will be entirely larger than 0.

$\Rightarrow (0.0534, 0.2466)$.